

Bhaskar Das

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EDUCATION

Concordia University

Montreal, QC

Bachelor of Engineering in Software Engineering

Expected Graduation: 05/2027

Relevant Coursework: Software Architecture and Design, Operating Systems, Database Systems, Computer Networks

EXPERIENCE

Reliance Jio Ltd.

05/2024 – 07/2024

Software Engineering Intern (Speech Systems & AI)

Mumbai, MH

- Engineered a **Speaker Diarization** pipeline using **NVIDIA TitaNet** and **Cosine Similarity** to identify speakers in **noisy audio**, enabling **high-confidence individual speaker captioning**.
- Applied **Signal Processing** and **MFCC** extraction to map acoustic signals to psychoacoustic scales, successfully **isolating vocal resonances** from background noise.
- Developed an **adaptive gain** script to apply a **7x increase** in signal scaling for speech segments identified via **VAD**, resulting in **high-confidence transcription** via **Whisper**.
- Standardized inconsistent webcam data by **resampling to 16kHz** and implementing a **500ms sliding window** to **balance precise speaker-switch detection** with system processing speed.

PROJECTS

AetherShell | *C, Swift (ARKit), Java (Spring Boot), Python, UDP/REST* github.com/das-bhaskar/aethershell

- Developed and maintain an open-source** bridge linking **legacy Nintendo DSi** hardware with modern **iOS ARKit** to project **2D sketches** as **3D animations**; featured on **Hackaday**.
- Published** a production **iOS App Store** release to display **AR-anchored animations** onto surfaces via the iPhone camera, driven by a **Python Vision Engine** for **2D rendering** and **Gemini API** classification.
- Resolved proxy-level integration** blockers and **HTTPS** limitations by injecting **custom HTTP headers** and offloading **SSL termination** to a **Java Hub**, allowing the **ARM9 CPU** to **bypass security landing pages**.
- Overcame** hardware networking constraints and **VRAM scan** latency by engineering a **UDP-based stylus telemetry** stream, achieving **real-time continuity** despite unstable, **legacy WiFi**.
- Engineered a fault-tolerant** pipeline using **Euclidean distance filtering** and vector math to maintain stroke integrity, effectively **eliminating coordinate jumps** caused by **ARM9 CPU** lag.
- Engineered a portable macOS deployment** and **short-hand QR handshake** to automate the **orchestration** of the **Spring Boot Hub**, **Python Vision Engine**, and **network tunnels** for a seamless **10-second setup** aimed at non-technical users.

ShareCycle | *Java, Spring Boot, MySQL, Docker, JUnit*

github.com/das-bhaskar/ShareCycle

- Collaborated** on a **microservices backend** using **Spring Boot**, implementing the **State Pattern** to manage **bike availability**, ensuring bikes in **Maintenance** could not be reserved to **eliminate booking conflicts**.
- Developed RESTful APIs** and **Stripe** integration with custom **bill breakdown logic**, automating **dynamic pricing** and financial processing for diverse bike categories.
- Containerized the multi-tier stack** using **Docker Compose**, achieving a **fully reproducible** development and deployment environment for the **MySQL** and API layers.
- Enforced** code quality via **JUnit testing**, achieving **65% coverage** to **validate payment edge cases** and ensure **accurate billing** across all rental tiers.

TECHNICAL SKILLS

Languages: Java, C, C++, Python, Swift, Kotlin, Bash

Frameworks & AI: Spring Boot, Flask, Streamlit, Android, iOS, JUnit, Scikit-learn, CUDA, NLP

Cloud & Databases: Supabase, Firebase, SQL (MySQL/PostgreSQL), NoSQL

Tools & DevOps: Git, Docker (Compose), Linux (Ubuntu/Debian), UNIX, GCC, Vim, CI/CD, GitHub Projects

Core Competencies: OOD/OOP, Design Patterns, Distributed Systems, Agile (SDLC), Data Structures & Algorithms, Network Protocols